
Shunting Inhibition and Dendritic Branching Shape Local Credit Assignment

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Abstract

1 Biological neurons assign credit through structures that standard neural networks
2 usually ignore: synapses are distributed across branching dendrites, inputs arrive as
3 excitatory and inhibitory conductances, and inhibition can shunt local voltage by
4 changing input resistance. We study conductance-based dendritic networks with E/I
5 synapse banks, shunting integration, and tree-structured branch-to-soma coupling,
6 and ask when a low-bandwidth somatic broadcast can replace the compartment-
7 specific errors of backpropagation. Exact gradients for this conductance tree show
8 that each synaptic update separates into a local eligibility term built from presyn-
9 aptic activity, driving force, and input resistance, and one non-local compartment
10 error. Under identity upward transfer, that error is the somatic error transported
11 through conductance-stage path gains. This factorization turns local learning into a
12 credit-signal compression problem: broadcast is useful only when it approximates
13 the path-specific compartment-error field. We test this mechanism across gradient,
14 oracle-transport, inhibition-intervention, competence, morphology, and harder-data
15 diagnostics. Shunting improves LocalCA-backprop alignment, reduces gradient-
16 scale mismatch, and makes exact compartment errors more rank-1 compressible
17 than matched additive controls in the regimes studied here. Inhibitory-conductance
18 interventions show that learned inhibition carries sample-specific pathway structure,
19 while transported-error oracle experiments show that restoring the exact path-gain
20 error largely closes the local-learning gap. Under nonnegative conductances and
21 rank-1 broadcast, shunting LocalCA remains within about 5–6 percentage points of
22 matched backpropagation reference performance on MNIST, Fashion-MNIST, and
23 context gating. These results suggest that E/I conductance, shunting inhibition, and
24 dendritic branching can reshape credit-signal geometry and make low-bandwidth
25 local learning more faithful.

26 1 Introduction

27 Unlike point units in standard neural networks, biological neurons assign credit across branching
28 dendritic trees. A synapse does not only see presynaptic activity and a scalar postsynaptic activation;
29 it also has access to local membrane voltage, synaptic reversal potential, total conductance, and
30 input resistance. Inhibitory input can therefore do more than subtract activity: by increasing local
31 conductance, it can shunt voltage and change the gain of every dendritic path passing through that
32 compartment.

33 We ask whether these biophysical ingredients can make local credit assignment easier. Backpropaga-
34 tion would assign a distinct error to every dendritic compartment, whereas a biologically plausible
35 supervisory signal is more likely to arrive as a low-bandwidth somatic or modulatory broadcast
36 [14]. The central question is therefore not whether such a broadcast can reproduce unconstrained

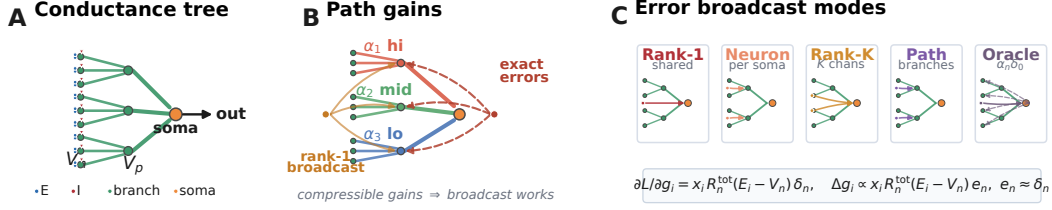


Figure 1: **Model and credit assignment.** (A) Conductance tree with nonnegative excitatory drive and inhibitory shunting load. (B) Exact credit varies by dendritic path, whereas rank-1 LocalCA broadcasts one shared error field. (C) LocalCA preserves the exact synapse-local eligibility and substitutes $e_n \approx \delta_n$; rank-1, neuron-wise, rank- K , and path-structured broadcasts trade biological bandwidth for fidelity, while transported error is an oracle.

backpropagation in every setting, but when the exact dendritic error field is simple enough for it to approximate. Our contribution is not simply to add dendritic structure to a neural network, but to show how conductance, shunting inhibition, and dendritic topology change the geometry of the credit signal itself.

Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for dendritic trees (Theorem 1). Each synaptic gradient factorizes into a synapse-local eligibility term and a single non-local compartment error. The eligibility term depends on presynaptic activity, driving force, and input resistance. The compartment error is path-specific: under identity upward transfer it is the somatic error multiplied by a conductance-stage path gain through the dendritic tree. LocalCA preserves the exact local eligibility and replaces this path-specific error with a broadcast field.

This factorization turns local credit assignment into a compressibility problem. By compressibility, we mean that the matrix of exact compartment errors across examples and compartments is well approximated by a low-rank signal, especially a rank-1 field shared across compartments. Dendritic branching creates the paths; E/I synapse banks set the local conductance state; and shunting inhibition changes the input resistance along each path. Shunting is not assumed to help monotonically. It helps rank-1 broadcast when inhibitory conductance attenuates high-gain paths or otherwise makes the exact error field more compressible.

Related work. This work builds on models of dendritic integration, branch-level nonlinear computation, normalization, and biologically plausible teaching signals [1, 3–6, 11–13, 16, 22–24, 30, 31, 44]. It also connects to machine-learning approaches that replace exact backpropagation with local, random, predictive, equilibrium, perturbation, or broadcast feedback signals [9, 10, 15, 25–29, 32–36]. Our contribution is complementary: we derive the exact conductance-tree credit object and test when shunting inhibition and dendritic morphology make that object broadcast-compatible.

We test this chain directly. Exact-gradient reconstruction verifies the factorization numerically. Gradient-fidelity diagnostics show that shunting produces local updates that are better aligned and better scaled relative to backpropagation than matched additive controls. Exact-error compressibility and inhibition-intervention diagnostics show that learned inhibitory conductance carries sample-specific pathway structure. Transported-error oracle experiments show that supplying the exact path-gain-transported compartment error largely closes the local-learning gap, identifying broadcast fidelity as the main bottleneck. In an MNIST gradient-dynamics diagnostic, shunting improves final local-backprop alignment (cosine 0.100 ± 0.033 vs. 0.005 ± 0.026) and reduces scale mismatch (0.26 ± 0.05 vs. 0.50 ± 0.05). On noise resilience under matched inhibition ($N_I \geq 5$), transported error raises additive local learning from 81–84% under rank-1 broadcast to 92–94%, nearly matching shunting at 95–95.2%.

Contributions. This paper makes three contributions. First, we derive the exact gradient for a conductance-based dendritic tree and show that synaptic credit separates into a local eligibility term and a path-specific compartment error. Second, we identify path-gain compressibility as the quantity that mediates whether a low-rank somatic broadcast can approximate this error. Third, we show empirically that shunting inhibition can improve this approximation by changing input resistance along dendritic paths, while also identifying regimes where rank-1 feedback fails and richer pathway-specific feedback is needed.

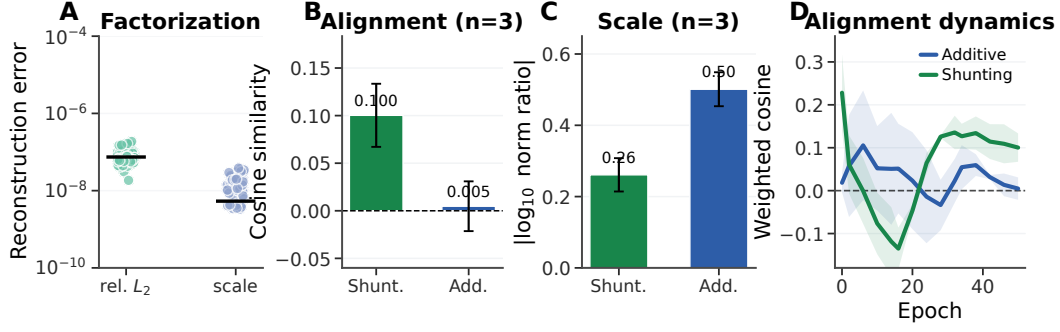


Figure 2: **Gradient-fidelity diagnostic.** (A) Exact-factorization sanity check: reconstructed gradients match autograd to numerical precision. (B) Final-state weighted cosine similarity between local and backprop gradients. (C) Final-state scale mismatch, $|\log_{10}(\|g^{\text{local}}\|/\|g^{\text{bp}}\|)|$. Error bars are ± 1 s.d. across 3 gradient-dynamics seeds; main learning and oracle sweeps use 5 seeds unless noted. (D) Alignment over learning; Appendix Fig. S3 shows finite gradient norms.

2 Compartmental Voltage Model and Gradient Derivation

We use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In normalized units, leak and inhibitory reversal are set to 0, the excitatory reversal is set to 1, and leak conductance is fixed at 1. Each compartment computes a conductance-weighted voltage: excitatory conductance pulls the voltage toward the excitatory reversal potential, leak and shunting inhibition pull it toward zero, and dendritic coupling transfers child voltages upward toward the soma. This form makes local sensitivity depend on both driving force ($E - V$) and total input resistance R^{tot} . Throughout the paper we write dendritic morphologies as rooted trees $[b_1, b_2, \dots, b_D]$, where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. Thus $[3, 3]$ means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for 9 distal leaves per soma. In the main feedforward experiments, synaptic inputs terminate on dendritic branches rather than directly on the soma. Each branch layer carries explicit excitatory conductance banks and, when enabled, learned inhibitory branch-level conductance banks driven by nonnegative inputs. This is input-driven shunting inhibition, not a recurrent lateral inhibitory circuit; explicit inhibitory-cell controls are reported in the appendix. The additive and shunting cores are architecture-matched at the tree level and differ only in the branch-level voltage integration rule. These ingredients play distinct roles in the credit-assignment argument. Branching creates the path structure. E/I synapse banks determine the local conductance state. Shunting turns inhibitory conductance into a multiplicative change in input resistance. The local learning rule then asks whether a low-bandwidth teaching signal can stand in for the exact path-specific error induced by that tree.

Notation. We use α_n for the exact conductance-stage path gain from compartment n to the soma, r_n^{4F} for the empirical 4F branch-soma covariance proxy, ϕ_n for the 5F confidence factor, λ_{mix} for the pathway-broadcast mixing weight, and κ_j for depth-dependent modulation in morphology extensions. All conductances and presynaptic activities in the main experimental setting are nonnegative; additive voltages may be signed comparators, but the conductance variables that define the shunting model are nonnegative.

2.1 Voltage Equation and Local Sensitivities

Consider compartment n with synaptic inputs j (activity x_j , reversal E_j , conductance $g_j^{\text{syn}} \geq 0$) and dendritic inputs from children (voltage V_j , conductance $g_j^{\text{den}} \geq 0$). The steady-state voltage is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \quad R_n^{\text{tot}} = 1/g_n^{\text{tot}}. \quad (1)$$

107 V_n is a convex combination of reversal potentials, child voltages, and leak. Writing $\mathcal{S}_n = \{0\} \cup$
 108 $\{E_j\}_j \cup \{V_j\}_j$, we have $\min \mathcal{S}_n \leq V_n \leq \max \mathcal{S}_n$ and $0 < R_n^{\text{tot}} \leq 1$. The local sensitivities follow
 109 directly:

Proposition 1 (Local Sensitivities).

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = x_i R_n^{\text{tot}} (E_i - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\text{den}} R_n^{\text{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\text{den}}} = R_n^{\text{tot}} (V_i - V_n). \quad (2)$$

110 Eq. (2) gives the eligibility factors used by the local rules below: presynaptic activity or voltage
 111 difference, driving force, and input resistance.

112 2.2 Shunting Inhibition as Divisive Gain Control

113 An inhibitory synapse with $E_{\text{inh}} \approx 0$ contributes current $(0 - V_n)x_j g_j^{\text{syn}}$ and increases g_n^{tot} . Its
 114 sensitivity is $\partial V_n / \partial g_j^{\text{syn}} = -x_j R_n^{\text{tot}} V_n$, which corresponds to multiplicative attenuation (divisive
 115 normalization). Shunting is divisive at the voltage level, but its effect on firing rates can be subtractive
 116 in some regimes [7]. Inhibitory plasticity can balance excitation dynamically [8]; our learned
 117 inhibitory conductances serve an analogous role. The same denominator also gives the direct
 118 connection to path gain. Writing $G_n^E(x)$ and $G_n^I(x)$ for the total excitatory and inhibitory synaptic
 119 conductance impinging on compartment n ,

$$R_n^{\text{tot}}(x) = (1 + G_n^E(x) + G_n^I(x) + G_n^{\text{den}})^{-1}.$$

120 Increasing $G_n^I(x)$ lowers $R_n^{\text{tot}}(x)$ and therefore multiplicatively suppresses every upstream path
 121 whose error must pass through compartment n . In this sense inhibition can gate credit flow by
 122 changing the path gain, even when the inhibitory drive is feedforward rather than lateral. Prop. 2
 123 formalizes this statement after the tree path gain is defined.

124 2.3 Exact Gradients for Dendritic Trees

125 Let V_0 be the somatic output and define the soma/core error as $\delta_0 := \partial L / \partial V_0$. For a linear decoder
 126 $\hat{y} = W_{\text{dec}} V_0$, this reduces to $\delta_0 = W_{\text{dec}}^\top (\partial L / \partial \hat{y})$; for a nonlinear decoder, δ_0 is the corresponding
 127 decoder-input Jacobian product. The theoretical derivation is stated for the conductance-stage
 128 voltage before any post-voltage reactivation. When a branch applies a monotone reactivation before
 129 transmitting activity upward, the same recursion holds with additional local derivative factors along
 130 the path. All exact-error diagnostics below are computed on pre-activation voltages, so they test the
 131 conductance-stage quantity derived here.

132 **Theorem 1** (Conductance-Stage Backpropagation on a Dendritic Tree). *For a rooted dendritic tree*
 133 *with soma at node 0 and unique parent $p(n)$ for each non-somatic compartment, suppose the parent*
 134 *receives either the pre-activation voltage V_n or a local post-voltage activity $a_n = f_n(V_n)$. For*
 135 *identity upward transfer, the loss gradient satisfies*

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}, \quad (3)$$

136 *with boundary condition $\partial L / \partial V_0 = \delta_0$. With post-voltage transfer $a_n = f_n(V_n)$, the local factor*
 137 *becomes $f'_n(V_n) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}$. Because the graph is a tree, each compartment has a unique path to*
 138 *the soma. Defining the conductance-stage path gain*

$$\alpha_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \alpha_n \delta_0, \quad (4)$$

139 *with $\alpha_0 = 1$ under identity transfer. In the post-voltage case, the exact effective path gain is obtained*
 140 *by multiplying the same product by the local factors $f'_i(V_i)$ on the transmitted child activities along*
 141 *the path.*

142 *Proof.* Apply the chain rule on the tree-structured computation graph using Prop. 1. $\square$

143 The one-step recursion in Eq. (3) and the path product in Eq. (4) separate local conductance transfer
 144 from the non-local somatic error.

Proposition 2 (Inhibitory control of conductance-stage path gain). *Let $\mathcal{A}(n)$ be the set of compartments on the path from compartment n to the soma, excluding n and including the parent compartments through which the error must pass. Holding dendritic coupling conductances fixed, an added inhibitory conductance $\Delta G_k^I(x) \geq 0$ at any $k \in \mathcal{A}(n)$ changes the conductance-stage path gain by*

$$\frac{\alpha_n(x; \Delta G^I)}{\alpha_n(x; 0)} = \prod_{k \in \mathcal{A}(n)} \frac{g_k^{\text{tot}}(x)}{g_k^{\text{tot}}(x) + \Delta G_k^I(x)}, \quad \frac{\partial \log \alpha_n}{\partial G_k^I} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathcal{A}(n), \\ 0, & k \notin \mathcal{A}(n). \end{cases} \quad (5)$$

Proof. Substitute $R_k^{\text{tot}} = 1/g_k^{\text{tot}}$ into the path-gain product in Eq. (4). Adding inhibitory conductance changes only the denominator at compartments on the path from n to the soma, which gives the product ratio and the log derivative in Eq. (5). $\square$

Prop. 2 makes two directed predictions. First, inhibition is a path gate: it attenuates all credit paths below the inhibited compartment. Second, inhibition is useful for rank-1 local learning only when this attenuation makes the distribution of α_n more uniform or better aligned with the task. If inhibition is too strong, too heterogeneous, or generated by a poorly calibrated inhibitory population, it can suppress useful signals and hurt learning. In the additive control the same I input changes voltages and local eligibilities, but it does not enter the conductance-stage path recursion as a multiplicative resistance gate.

When does shunting compress path gains? The attenuation identity above is not by itself a concentration result. Let $z_n = \log \alpha_n$ be the log path gain for compartment n . If added inhibitory conductance contributes a path-dependent attenuation

$$\beta_n(x) = \sum_{k \in \mathcal{A}(n)} \log \left(1 + \frac{\Delta G_k^I(x)}{g_k^{\text{tot}}(x)} \right), \quad z'_n = z_n - \beta_n, \quad (6)$$

then, over compartments or examples,

$$\text{Var}(z') = \text{Var}(z) + \text{Var}(\beta) - 2 \text{Cov}(z, \beta). \quad (7)$$

Thus shunting narrows the log path-gain field exactly when

$$\text{Cov}(z, \beta) > \frac{1}{2} \text{Var}(\beta). \quad (8)$$

Eq. (6) follows by taking the logarithm of the product ratio in Prop. 2; Eq. (7) then applies the variance identity for $z - \beta$. Eq. (8) is an interpretive sufficient condition under the same averaging measure, not a primary empirical diagnostic. In finite trained networks, the covariance margin can disagree with CV and exact-error-rank summaries because these statistics average over different compartment and sample axes. We therefore use direct path-gain dispersion, exact-error compressibility, and inhibition-intervention measurements as the empirical tests, and report the covariance margin as a boundary check in the appendix.

Corollary 1 (Local-Global Factorization). *The exact synaptic gradient at compartment n factorizes as:*

$$\frac{\partial L}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\text{compartment error}}, \quad (9)$$

where the eligibility term depends only on quantities available at synapse i (presynaptic activity x_i , input resistance R_n^{tot} , driving force $E_i - V_n$), and the compartment error $\partial L / \partial V_n$ is the sole non-local quantity.

The factorization in Eq. (9) isolates the only non-local quantity as $\partial L / \partial V_n = \alpha_n \delta_0$ under identity upward transfer. With post-voltage transfer, the activation-derivative factors in Theorem 1 enter the effective path gain, but the compartment error remains the only non-local term. Any practical local rule therefore depends entirely on how well a broadcast signal approximates that exact compartment error, which we evaluate in Sec. 4.2. In the models studied here, shunting improves this approximation by reducing and stabilizing R_n^{tot} , which narrows the spread of local sensitivities across the tree. In the empirical networks, each dendritic branch layer may optionally apply a post-voltage reactivation $f(V_n)$ before passing activity to the next stage; unless otherwise noted, the manuscript sweeps use a learnable bounded `param_tanh` reactivation.

3 Local Learning Rules

3.1 Broadcast Error Approximation

We approximate the exact compartment error $\partial L/\partial V_n$ (Corollary 1) with a broadcast signal e_n derived from the somatic/core error δ_0 . For a linear readout this reduces to $\delta_0 = W_{\text{dec}}^\top (\partial L/\partial \hat{y})$; for a nonlinear decoder we use the corresponding decoder-input Jacobian product $J_{\text{dec}}(h_{\text{core}})^\top (\partial L/\partial \hat{y})$ so that the broadcast remains defined in the same core-space coordinates. We distinguish five feedback objects:

- **Rank-1 shared broadcast:** $e_n = \bar{\delta} \cdot \mathbf{1}$, where $\bar{\delta} = \text{mean}(\delta_0)$ is one scalar field delivered to all compartments.
- **Neuron-wise broadcast:** $e_n = \delta_0$ when the layer width matches the core/output error dimension. In our main feedforward settings this condition usually fails, so the default neuron-wise implementation reduces to rank-1 shared broadcast.
- **Rank- K random broadcast:** $c = P_K \delta_0 \in \mathbb{R}^K$ and $e_n = \sum_{k=1}^K q_{n,k} c_k$, where P_K is a fixed random projection and $q_{n,k}$ mixes those K channels into compartment n .
- **Path-structured broadcast:** $e_n = \lambda_{\text{mix}} \bar{\delta} \cdot \mathbf{1} + (1 - \lambda_{\text{mix}}) \Gamma_n(\delta_0)$, where Γ_n projects the somatic error into router-inferred pathway roles, gates those roles by local pathway activity, and transports the resulting vector hierarchically through the block-structured dendritic tree [43].
- **Transported oracle:** $e_n^{\text{transport}} = \alpha_n \delta_0 = \partial L/\partial V_n$. This is an analysis upper bound, not a proposed local rule, because it supplies the exact path gain α_n from Theorem 1.

In the main feedforward architectures, hidden widths exceed the decoder-error dimension, so the default neuron-wise broadcast reduces to a rank-1 shared field (Appendix Table S11). Thus the main rank-1 result should be read as the architecture-induced low-bandwidth case, not as a claim that neuron-wise broadcasts were unavailable in principle. Under shunting, that deliberately low-bandwidth field is often sufficient on standard classification tasks. Routed tasks and rank-bridge controls in the appendix test the opposite regime, where rank-1 broadcast collapses branch identity and richer feedback is needed. A local-mismatch heuristic is included only as an appendix control because it performs much worse than the somatic-error broadcasts (Appendix A).

3.2 Three-Factor Rule (3F)

Definition 1 (3F Update). *For synaptic and dendritic conductances:*

$$\Delta g_j^{\text{syn}} = \eta \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B, \quad \Delta g_j^{\text{den}} = \eta \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B, \quad (10)$$

where $\langle \cdot \rangle_B$ denotes the batch average.

The three factors are presynaptic activity x_j (or a voltage difference), postsynaptic modulation through driving force and input resistance, and the broadcast error e_n . The same rule applies to excitatory and inhibitory synapses. The sign difference comes from the driving force ($E_j - V_n$). We write Δg as the local gradient estimate supplied to the optimizer; the optimizer applies the usual descent step. Equivalently, one could absorb the minus sign into e_n and treat it as a descent signal.

Additive control. Rule (10) is the local gradient of the *shunting* voltage $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$. For the additive control ($V_n = \sum g_j x_j$, no divisive normalization), the correct local gradient is simpler: $\Delta g_j^{\text{syn}} = \eta \langle x_j e_n \rangle_B$, with no driving-force or R_n^{tot} terms ($R_n^{\text{tot}} \equiv 1$ by definition since there is no denominator). Throughout, each architecture uses the learning rule derived from its own forward-pass dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived rule to an additive model.

3.3 Practical Heuristic Wrappers: 4F and 5F

4F (branch-soma relevance proxy). In the conductance-stage recursion (4), the path gain α_n attenuates the somatic error differently at each compartment. To compensate without explicitly

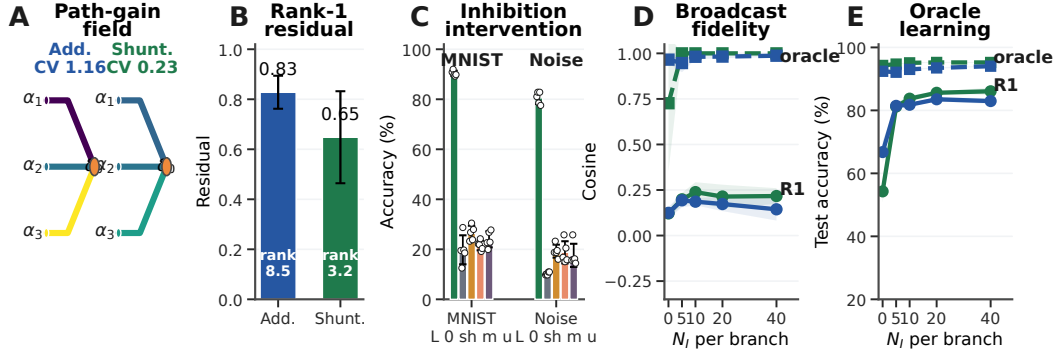


Figure 3: **Mechanistic chain from path gains to learning.** (A) Dendritic path-gain fields at matched inhibition. (B) Exact-error compressibility on MNIST: lower rank-1 residual and participation rank favor scalar broadcast. (C) Post-training inhibitory-conductance interventions (L: learned; 0/sh/m/u: zero, shuffled, mean-clamped, uniform matched). (D) Compartment-error fidelity on noise resilience; dashed lines show the transported-error oracle. (E) Transported-error learning recovers near-ceiling local learning where rank-1 broadcast fails. Error bars denote ± 1 s.d. across seeds.

231 computing that gain, 4F uses an online correlation proxy

$$r_n^{4F} = \text{Cov}(\bar{V}_n, \bar{V}_0) / (\sqrt{\text{Var}(\bar{V}_n) \text{Var}(\bar{V}_0)} + \varepsilon).$$

232 High r_n^{4F} up-weights branches whose voltages covary with the soma; low r_n^{4F} down-weights branches
 233 where the broadcast is likely to be a poor proxy. We keep 4F as a diagnostic intermediate. Empirically,
 234 this factor alone gives little improvement over 3F; the practical jump comes from the 5F confidence
 235 factor below.

236 **5F (confidence-weighted 4F).** 5F adds a second heuristic factor

$$\phi_n = \text{Var}(V_n) / (\sigma_{\text{res}}^2 + \varepsilon),$$

237 where σ_{res}^2 is the residual variance of V_n after linear regression on the parent voltage. Operationally,
 238 ϕ_n is a branch-wise confidence gate, clamped to $[0.25, 4.0]$ for stability. This factor is empirical, not
 239 theorem-derived; its gain should be read as practical denoising rather than theoretical necessity.

Definition 2 (5F Update).

$$\Delta g_j^{\text{syn}} = \eta r_n^{4F} \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B. \quad (11)$$

240 Eq. (11) is therefore the main practical LocalCA update, while the 3F rule is the theoretically derived
 241 local gradient and 4F is the intermediate ablation. A feedback-alignment-style random-broadcast
 242 argument is included in Appendix C; it is supportive but not central to the main mechanism claim.

243 4 Experiments

244 4.1 Setup

245 The experiments test five linked predictions: exact factorization should reconstruct autograd gradients;
 246 shunting should improve LocalCA direction and scale relative to backpropagation; exact compartment
 247 errors should be more compressible under shunting than under matched additive controls; learned
 248 inhibition should carry sample-specific pathway structure rather than merely adding conductance load;
 249 and replacing the broadcast with transported exact error should largely close the local-learning gap.
 250 We evaluate capacity-calibrated supervised tasks and controlled sweeps that isolate inhibition strength,
 251 rule family, broadcast mode, morphology, and architecture. The main text uses MNIST, Fashion-
 252 MNIST [37], context gating, noise resilience, and a compressed morphology-regime summary;
 253 CIFAR-10 and routed-feedback diagnostics are reported in the appendix (Figs. S7, S11). Main
 254 comparisons are architecture-matched additive vs. shunting dendritic cores and local vs. backprop
 255 training; DFA is a supplementary baseline. Unless otherwise stated, local runs use 5F with the default

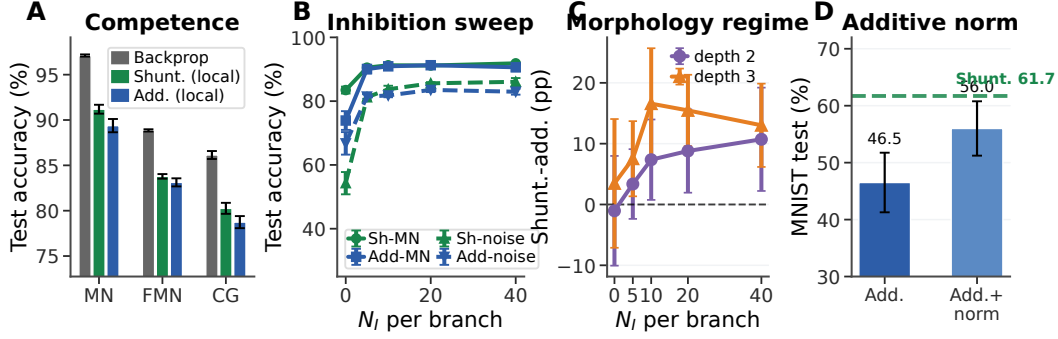


Figure 4: **Capacity-calibrated competence and regime dependence.** (A) Matched backprop reference performance vs. rank-1 5F LocalCA on MNIST, Fashion-MNIST, and context gating. (B) Inhibitory dose-response. (C) Morphology-dependent shunting advantage on noise resilience from a 3-seed sweep. (D) Additive normalization improves additive LocalCA but remains below the shunting reference. Error bars denote ± 1 s.d. where available.

neuron-wise broadcast, which reduces to rank-1 shared broadcast when hidden width exceeds the decoder-error dimension. All main-claim dendritic runs use nonnegative conductance transforms and nonnegative first-layer synaptic drive; image inputs are in $[0, 1]$, corrupted inputs are clipped back to $[0, 1]$, and context gating uses a nonnegative contextual figure-ground MNIST construction. The transported-error condition is an oracle upper bound derived from Theorem 1, not the proposed biological rule. Main mechanistic, oracle, and competence results use 5 seeds per condition unless noted otherwise.

4.2 From Exact Factorization to Credit-Signal Fidelity

To test whether these performance differences correspond to better credit signals, we compare local and backprop gradients on the same batch at the same parameter values. For each parameter tensor p , we compute directional alignment (cosine similarity) and scale mismatch, defined as $|\log_{10}(\|g_p^{\text{local}}\|/\|g_p^{\text{bp}}\|)|$, aggregated by parameter count.

Across the MNIST gradient-dynamics diagnostic, shunting is more aligned with backprop and closer in norm than the additive control at the final checkpoint: weighted cosine rises from 0.005 ± 0.026 to 0.100 ± 0.033 , and scale mismatch falls from 0.50 ± 0.05 to 0.26 ± 0.05 . Figure 2A confirms that the implementation matches the theory under the nonnegative-input setup: reconstructing gradients from Corollary 1 using exact compartment errors yields weighted relative reconstruction error below 2×10^{-7} and weighted scale mismatch below 4×10^{-8} across runs. In a few low-norm parameter blocks, cosine becomes numerically unstable even though the reconstruction itself is essentially exact, so we report the norm-based diagnostics in the figure. Figure 2D shows the same gradient-alignment comparison over training: additive alignment remains near zero at the final checkpoint rather than failing only at one static snapshot. Appendix Fig. S3 verifies that the corresponding local and backprop gradient norms remain finite throughout learning, ruling out a trivial zero-gradient explanation. At the level of the theoretically derived 3F rule, the central non-local approximation is therefore the broadcast substitute for $\partial L/\partial V_n$.

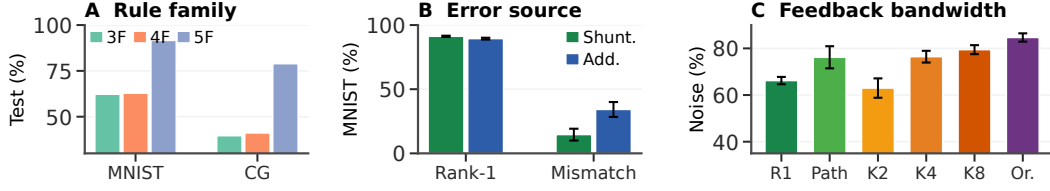


Figure 5: **Rule and feedback controls.** (A) The practical gain comes mainly from the 5F confidence wrapper; 3F/4F remain weak optimizers. (B) Replacing the somatic rank-1 error with a local-mismatch heuristic collapses learning. (C) Richer or transported feedback improves noise resilience. Detailed controls remain in Appendix Fig. S1.

4.3 Mechanistic Chain: Path Gains, Broadcast Fidelity, and Oracle Transport

Figure 3 provides the central empirical chain. At matched inhibition ($N_I=5$), shunting narrows the MNIST conductance-stage path-gain field (CV 0.23 vs. 1.16 for additive) and makes the exact compartment-error matrix more compressible: the best rank-1 residual drops from 0.83 ± 0.07 to 0.65 ± 0.18 , while participation rank drops from 8.5 ± 3.3 to 3.2 ± 1.2 . This is the diagnostic matched to the paper’s main claim: rank-1 broadcast is useful when the exact error field is closer to low rank.

The inhibition intervention shows that the learned inhibitory field is necessary for the trained computation and is not interchangeable with a matched conductance load: zeroing, shuffling, or clamping learned branch inhibition drops MNIST from about 90% to 20–26% and noise resilience from about 81% to 10–19%. The fidelity and oracle panels test the same bottleneck: rank-1 broadcast tracks the exact compartment error better under shunting once inhibition is present, transported error reaches about 0.95–1.0 fidelity, and transported learning lifts both cores to 92–95% on noise resilience. CIFAR-10 and direct-I versus explicit-I path-gain probes remain appendix diagnostics (Fig. S7; Table S4).

4.4 Capacity-Calibrated Competence

In the capacity-calibrated regime, shunting LocalCA remains within about 5–6 percentage points of matched backpropagation on MNIST, Fashion-MNIST, and context gating (Table S2; Fig. 4). Figure 4D shows that additive normalization improves additive LocalCA but remains below the shunting reference, so the gain is not explained by generic voltage normalization alone. Figure 5 summarizes the rule and feedback controls: 5F is the strongest practical optimizer, somatic rank-1 error is essential, and higher feedback bandwidth improves the noise-resilience diagnostic.

4.5 Regime Dependence of the Shunting Advantage

With matched rank-1 broadcast, shunting is modestly better on MNIST, Fashion-MNIST, and context gating, including 0.803 ± 0.006 vs. 0.787 ± 0.007 for additive on context gating. The gap grows on inhibition-sensitive noise resilience once inhibitory conductance is present (Fig. 4B). Morphology shifts the useful inhibitory operating regime: different tree depths and branch factors peak at different N_I , and the shunting advantage is not monotonic in inhibition (Fig. 4C). Appendix Fig. S5 extends this boundary check to depth scaling and noisy error broadcasts.

5 Discussion and Limitations

We studied local credit assignment in dendritic networks that explicitly represent three biological ingredients often abstracted away in standard neural networks: E/I conductance synapses, shunting inhibition, and tree-structured dendritic branching. The exact gradient derivation shows that these ingredients create a simple decomposition. The synapse-local part of the gradient is available from presynaptic activity, driving force, and input resistance; the only non-local quantity is a compartment-specific error.

The experiments support a coherent mechanism. Branching creates multiple credit paths from distal synapses to the soma, E/I synapse banks set the conductance state of each compartment, and shunting

inhibition modulates input resistance along those paths. In the regimes studied here, this makes the exact error field more compressible for rank-1 broadcast, improves local-backprop gradient alignment, and reduces scale mismatch. Transported-error oracle experiments show that replacing the broadcast with the exact path-gain-transported error largely closes the local-learning gap, while inhibition interventions show that learned inhibitory conductance carries sample-specific pathway structure rather than merely setting a global conductance scale.

The mechanism has clear limits. Shunting is not a universal substitute for backpropagation, and more inhibition is not always better. The practical 5F rule is an empirical stabilizer rather than a theorem-derived necessity; Eq. (8) is interpretive; activation shape changes the operating regime; and routed or harder-data settings require higher-rank or pathway-specific feedback when the exact error field is not rank-1 compressible. The main model uses input-driven shunting inhibition rather than recurrent lateral inhibition, so recurrent inhibitory circuits remain a future direction. The main message is that biophysical dendritic mechanisms can change not only neural representations, but also the geometry of credit assignment.

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A Supplementary Results

Appendix overview. The appendix is organized by how directly each result supports the main mechanism. The first block collects rule and feedback controls that complement Fig. 5. The second block reports competence checks, gradient dynamics, stress tests, and baseline comparisons. The third block contains harder-data, morphology, inhibition, and pathway-feedback diagnostics. The final blocks give implementation, theory, task, and concise secondary weight statistics. Exploratory screens and development summaries that do not directly support a paper claim are omitted.

Boundary diagnostics. Several controls are intentionally negative or mixed. The log-gain covariance margin from Eq. (8) is not a positive main result, Double-CIFAR remains near chance even under matched backpropagation, positive cue-routing controls saturate too strongly to separate mechanisms, and the current pathway-vector implementation does not beat the best unstructured low-rank cue-routing control. We include these results to delimit the claim rather than to smooth over failure modes.

Mechanistic Support and Broadcast Controls

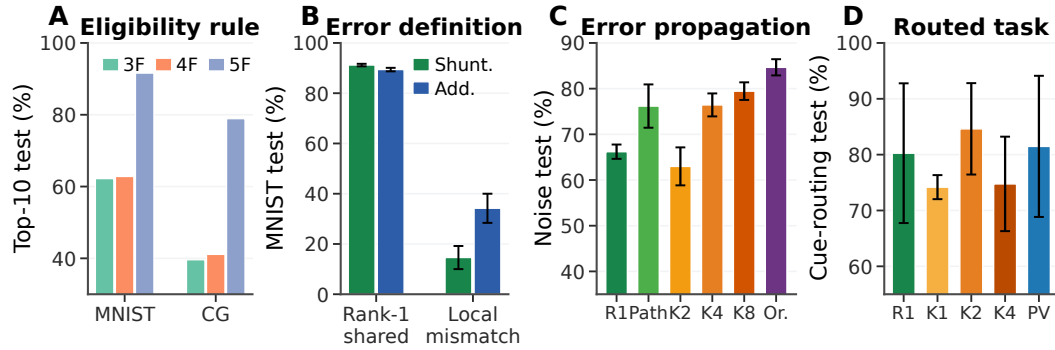


Figure S1: **Local-rule and feedback-design axes.** (A) The theoretically derived 3F update and the 4F attenuation proxy are weak practical optimizers in the completed local-competence sweeps; 5F is the strongest empirical wrapper. (B) The broadcast definition matters: on MNIST, rank-1 shared somatic error is strong, while a local-mismatch error surrogate fails. (C) On noise resilience, recursively propagating or increasing the rank of the error field closes much of the gap between rank-1 broadcast and exact path transport. (D) Cue routing probes a branch-specific regime: higher rank helps, while the current pathway-vector implementation is useful but not yet better than the best unstructured low-rank control. Error bars denote ± 1 s.d. where five-seed summaries are available.

Rule-Family Ranking

Dataset	Rule	Top-10 valid	Top-10 test
MNIST	3F	0.611	0.622
MNIST	4F	0.620	0.628
MNIST	5F	0.912	0.916
Context gating	3F	0.398	0.396
Context gating	4F	0.411	0.411
Context gating	5F	0.807	0.789

Table S1: **Rule-family ranking.** Top-10 mean across completed local-competence sweeps.

Dataset	BP ceiling	Shunting local	Additive local	BP gap
MNIST	0.971	0.912 ± 0.005	0.894 ± 0.007	5.9%
Fashion-MNIST	0.889	0.838 ± 0.003	0.831 ± 0.004	5.1%
Context gating	0.861	0.803 ± 0.006	0.787 ± 0.007	5.8%

Table S2: **Local competence.** Backprop ceilings are matched-capacity shunting references from dedicated 5-seed standard-training refreshes for MNIST and context gating and from the dedicated 5-seed Fashion-MNIST competence sweep. Local values are 5F with rank-1 broadcast and local decoder updates. Context gating additionally uses an HSIC auxiliary objective (weight 0.01; Appendix C). Error bars on local values: ± 1 s.d. across 5 seeds.

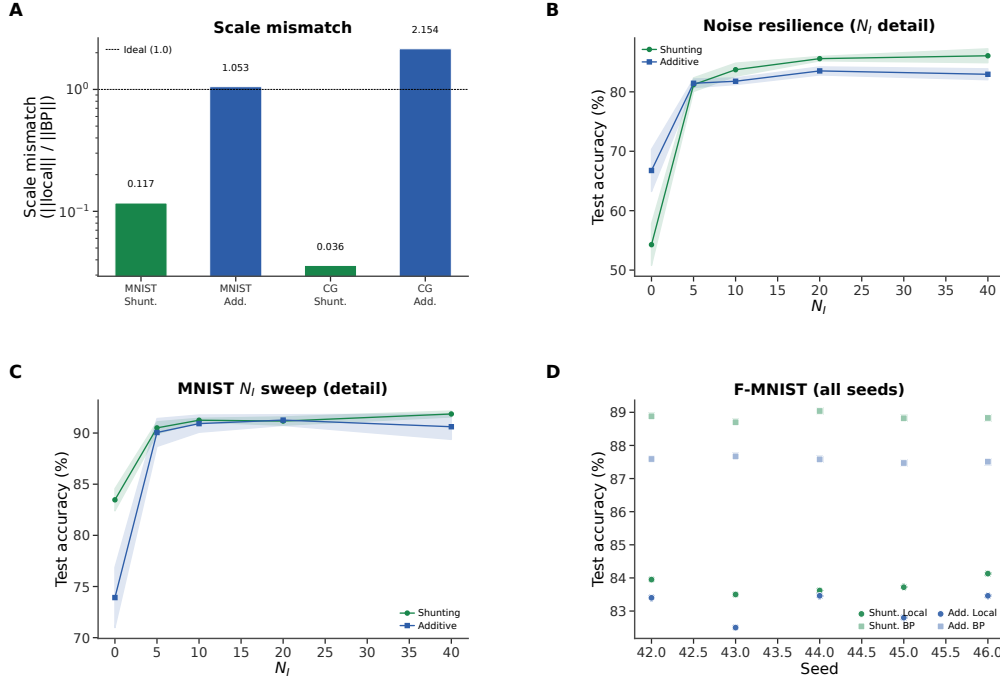


Figure S2: **Extended gradient and N_I analysis (supplementary).** (A) Scale mismatch bars from a separate matched-weight static diagnostic; these values should not be compared directly to the checkpointed final-state diagnostic in Fig. 2. Shunting achieves near-ideal scale (0.117); additive exhibits order-of-magnitude distortion (> 1.0). (B) Noise-resilience N_I dose-response with error bands (± 1 s.d.). (C) MNIST N_I dose-response detail with error bands. (D) Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

442 Competence, Verification, and Stress Tests

443 Local Competence and Regime Dependence

444 Extended Gradient Analysis and N_I Sweep Detail

445 **Alignment dynamics are not a zero-gradient artifact.** To check whether low additive alignment
446 could be caused by vanishing gradients, we analyzed a separate 3-seed checkpoint trajectory with
447 parameter-level LocalCA and backprop gradients. The additive local gradients are nonzero throughout
448 training (weighted norm range 9.5×10^{-4} – 9.5×10^{-3}), and the corresponding backprop gradients are
449 also nonzero (2.3×10^{-5} – 5.7×10^{-3}). The low additive cosine therefore reflects directional and scale
450 mismatch, not absent gradients: additive 5F cosine stays near zero from epoch 0 to epoch 50 (0.018
451 to 0.005), while its local/backprop norm ratio falls from about 8.8×10^2 to 0.69.

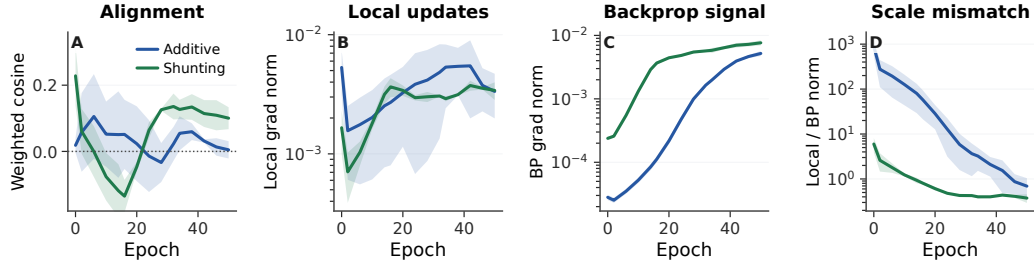


Figure S3: **Alignment dynamics with explicit gradient-norm checks.** (A) Weighted LocalCA-vs.-backprop cosine over training for the 5F rule. (B,C) Weighted local and backprop gradient norms stay nonzero for both additive and shunting models. (D) The additive control begins with a much larger scale mismatch, then approaches the backprop scale without recovering strong directional alignment. Shading is s.d. across 3 seeds.

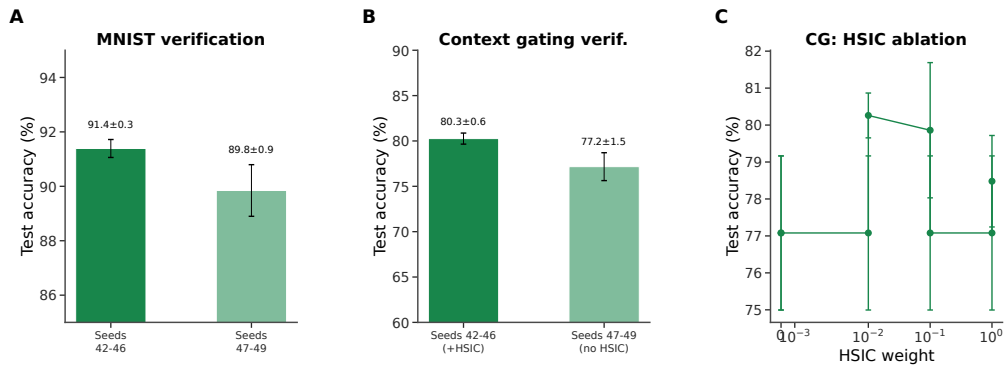


Figure S4: **Verification and reproducibility (supplementary).** (A) MNIST verification: main seeds (42–46) yield $91.4 \pm 0.3\%$; held-out seeds (47–49) yield $89.8 \pm 0.9\%$, confirming generalization. (B) Context gating verification: main seeds (+HSIC) $80.3 \pm 0.6\%$; held-out seeds (no HSIC) $77.2 \pm 1.5\%$. The roughly 3 pp gap reflects HSIC removal, not seed sensitivity. (C) HSIC weight ablation on context gating: moderate weights (0.01–0.1) perform best.

452 Verification and Reproducibility

453 Additional Stress Tests

454 FA/DFA Baseline Comparison

455 CIFAR-10 Results

456 Table S3 provides the exact values for the appendix CIFAR-10 figure. In this stronger compact direct-
 457 I-stream family, shunting performs well under standard training (49.5% vs. 48.3% for additive), so
 458 the harder-data stress test does not unfairly penalize the shunting architecture. The broadcast-fidelity
 459 ladder also survives once decoder-aware soma mapping is used: additive LocalCA reaches 25.5%
 460 under rank-1 per-soma broadcast and 32.4% under transported error, while shunting rises from 29.7%
 461 under rank-1 broadcast to 35.0% with random low-rank $K=4$ and 47.2% with transported error,
 462 nearly matching the 49.5% shunting backprop ceiling. Removing learned I-to-E conductance lowers
 463 the shunting ceiling by 5.7 pp and transported LocalCA by 8.6 pp, making this the clearest harder-data
 464 evidence that inhibitory conductance is part of the useful credit geometry rather than just an added
 465 architectural detail.

466 A routed Double-CIFAR follow-up with strictly nonnegative router outputs was also not included as
 467 positive evidence. The task remained near chance even under matched backprop: additive BP reached
 468 $10.9 \pm 0.7\%$, shunting BP with learned I-to-E reached $11.6 \pm 0.4\%$, and the strongest LocalCA
 469 condition, shunting transported error with learned I-to-E, reached only $11.7 \pm 0.3\%$. We therefore

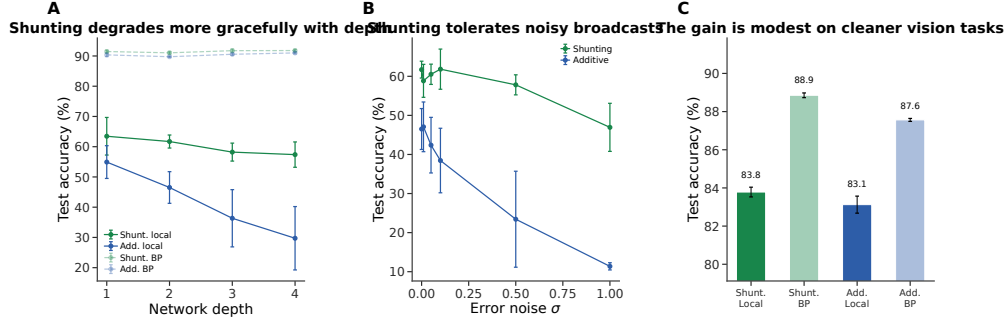


Figure S5: **Additional stress tests.** (A) Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Backprop ceilings (dashed) remain stable. (B) Broadcast-noise robustness: shunting remains stable for $\sigma \leq 0.1$; additive degrades rapidly and reaches chance at $\sigma = 1$. (C) Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (83.8% local vs. 88.9% backprop), consistent with the regime-dependent story in the main text.

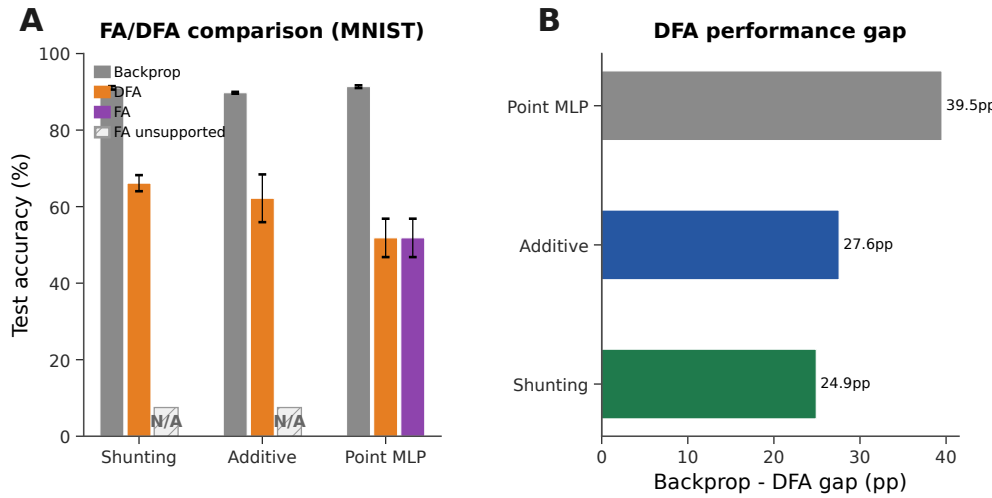


Figure S6: **Feedback alignment baselines (MNIST).** (A) Grouped comparison: standard backprop (neutral gray), DFA (orange), and FA (purple). Hatched N/A markers indicate that FA is not implemented for the block-structured dendritic architectures because the random feedback matrices are not dimensionally compatible with the tree-structured branch updates. DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. (B) Backprop–DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

470 treat this as a failed hard routed-task screen and keep the stronger compact CIFAR-10 control ladder
 471 as the paper's harder-data diagnostic.

472 **Additive Normalization Control**

473 **Dendritic Morphology, Pathway Structure, and Supportive Extensions**

474 **Morphology $\times$ Inhibition Regime Map**

475 This supportive regime map tests whether tree geometry changes the inhibitory operating point at
 476 which shunting helps. We treat it as evidence that morphology changes the useful inhibition regime;
 477 a fuller mechanistic account would still require matched theory diagnostics for each cell in the grid.

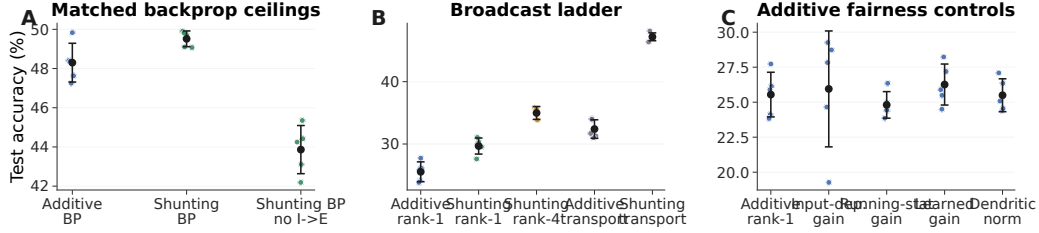


Figure S7: **CIFAR-10 control ladder in the compact direct-I-stream family.** (A) Matched backprop ceilings. With nonnegative inputs and a nonlinear decoder, shunting slightly exceeds additive under standard training, and removing learned I-to-E conductance lowers the shunting ceiling. (B) Broadcast ladder. Rank-1 broadcast remains weak on this harder dataset, rank-4 improves partially, and exact transported error nearly reaches the shunting backprop ceiling. (C) Additive fairness controls. Input-dependent gain, running-stat gain, learned gain, and dendritic normalization do not rescue additive rank-1 broadcast. This figure is a harder-data diagnostic, not a claim of competitive CIFAR-10 benchmarking.

Condition	Core	Training / broadcast	I-to-E	Test acc.
standard	additive	backprop	yes	48.3 $\pm$ 1.0
standard	shunting	backprop	yes	49.5 $\pm$ 0.4
standard, no learned inhibition	shunting	backprop	no	43.9 $\pm$ 1.2
rank-1	additive	5F per-soma	yes	25.5 $\pm$ 1.6
rank-1, input-dependent gain	additive	5F per-soma	yes	25.9 $\pm$ 4.1
rank-1, running-stat gain	additive	5F per-soma	yes	24.8 $\pm$ 0.9
rank-1, learned gain	additive	5F per-soma	yes	26.3 $\pm$ 1.5
rank-1, dendritic normalization	additive	5F per-soma	yes	25.5 $\pm$ 1.2
rank-1	shunting	5F per-soma	yes	29.7 $\pm$ 1.3
rank-1, no learned inhibition	shunting	5F per-soma	no	22.3 $\pm$ 0.7
transported error	additive	5F path transport	yes	32.4 $\pm$ 1.5
rank-4	shunting	5F low-rank	yes	35.0 $\pm$ 1.0
transported error	shunting	5F path transport	yes	47.2 $\pm$ 0.6
transported error, no learned inhibition	shunting	5F path transport	no	38.6 $\pm$ 1.0

Table S3: **Exact CIFAR-10 values for the 70-run control ladder (5 seeds per condition, validation-best epoch).** All rows use flattened CIFAR-10 in $[0, 1]$, nonnegative transfer outputs, positive conductance transforms, a compact depth-4 dendritic core, and a nonlinear decoder. LocalCA maps output error back to decoder-input/soma coordinates through the decoder Jacobian. The useful conclusions are narrow but important: learned shunting slightly improves the matched backprop ceiling, learned I-to-E conductance is necessary for the strongest shunting results, transported error nearly closes the shunting LocalCA gap, and additive rank-1 fairness controls do not explain the transported-shunting result.

478 Input-Mode Probe: Direct Inhibitory Stream vs. Explicit Inhibitory Cells

479 The main experiments use input-driven inhibitory streams: the transfer layer provides nonnegative
480 excitatory and inhibitory channels from the external input, and excitatory dendrites receive learned
481 I-to-E conductance banks directly. To test whether the same mechanism can be generated by an
482 explicit feedforward inhibitory population, we ran a one-feedforward-layer noise-resilience probe
483 that preserves the relevant dendritic depth ($[3, 3]$ morphology). In this clean setting, explicit inhibitory
484 cells reach $94.2 \pm 0.2\%$ under path-transport LocalCA when their dendrites use the same update policy
485 as excitatory dendrites, $93.8 \pm 0.2\%$ when those inhibitory-cell updates are frozen, and $95.1 \pm 0.3\%$
486 under standard backprop. Direct input-driven inhibition gives the expected broadcast-fidelity ladder
487 on the same one-layer architecture: rank-1 broadcast reaches $85.2 \pm 0.7\%$, while exact path transport
488 reaches $95.2 \pm 0.0\%$. Thus explicit inhibitory cells can generate useful path-gain structure once
489 the architecture isolates the dendritic-path question. The older two-feedforward-layer explicit-I
490 diagnostic was much weaker (about 58–63% under LocalCA despite a 94.7% backprop ceiling), so
491 we treat it as evidence that deeper explicit $E \rightarrow I \rightarrow E$ graphs need graph-aware local errors, not as
492 evidence against inhibition-mediated path gains.

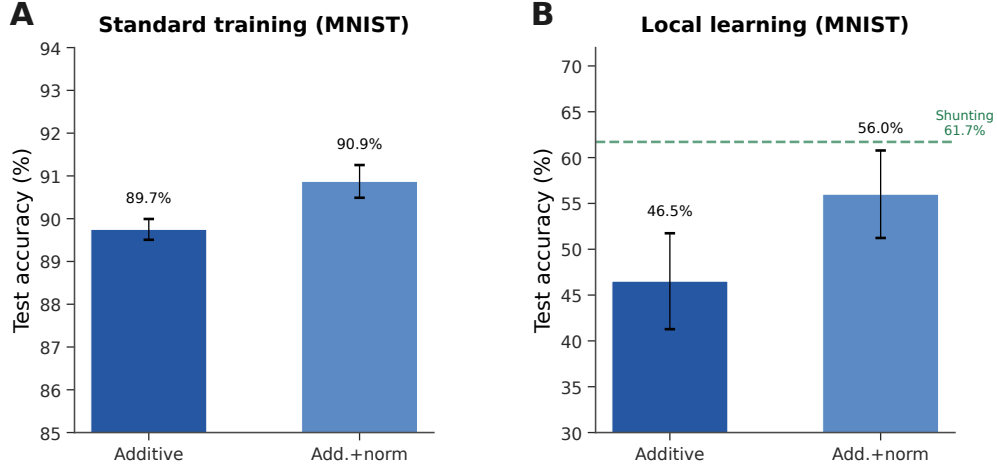


Figure S8: **Additive + normalization control (MNIST).** (A) Standard backprop: normalization provides a small boost (89.7% $\rightarrow$ 90.9%). (B) Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). The remaining gap (-5.7 pp) indicates that shunting provides benefits beyond simple voltage normalization.

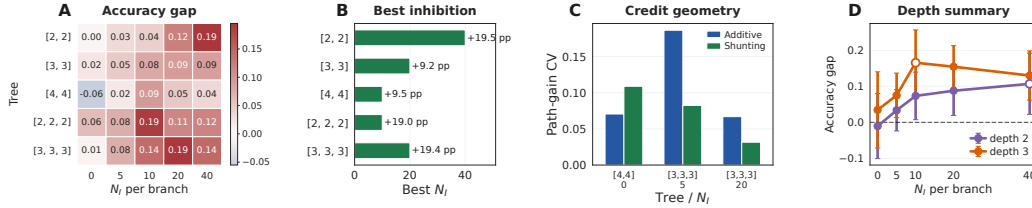


Figure S9: **Morphology shifts the useful inhibitory regime.** (A) Shunting-minus-additive LocalCA accuracy gap across dendritic morphologies and inhibitory synapse counts. (B) Best inhibitory count for each morphology. (C) Representative path-gain diagnostic for selected morphology cells, used to connect the performance map to credit geometry. (D) Average shunting advantage by depth, with markers showing the peak inhibitory count for each depth.

493 **Post-training inhibition intervention and exact-error rank.** We next tested whether learned
 494 inhibitory conductance carries task-specific structure rather than only changing global scale. We
 495 replayed trained shunting LocalCA checkpoints while replacing the branch-level inhibitory conduc-
 496 tance with four post-training interventions: zero inhibition, sample-shuffled inhibition, per-branch
 497 batch means, or one uniform matched mean. On a 512-example held-out subset, learned MNIST
 498 inhibition at $N_I=5$ gives $90.4 \pm 1.1\%$ accuracy, while the four interventions drop to 19.8–26.4%.
 499 On noise resilience at $N_I=10$, learned inhibition gives $80.5 \pm 2.4\%$, while the interventions drop
 500 to 10.3–19.3%. Thus the inhibitory field is not interchangeable with a matched global conductance
 501 load. We also computed the SVD of exact compartment-error matrices across samples and compart-
 502 ments. At matched MNIST $N_I=5$, shunting has lower rank-1 residual than additive (0.65 ± 0.18
 503 vs. 0.83 ± 0.07) and lower participation rank (3.2 ± 1.2 vs. 8.5 ± 3.3), supporting the path-gain
 504 compressibility interpretation.

505 We also computed the direct log-gain covariance margin in Eq. (8) on the selected morphology
 506 diagnostic checkpoints. The margin is zero when no inhibitory conductance is present and slightly
 507 negative in the inhibited shunting selections (-2.5×10^{-3} , -5.0×10^{-4} , and -2.8×10^{-4} at
 508 $N_I=5, 20, 40$). We therefore do not use this covariance condition as a positive main result. Its
 509 role is to clarify when shunting should compress gains; the empirical claim rests on the measured
 510 comparative path-gain dispersion, exact-error rank, and inhibition-intervention diagnostics.

Inhibitory input mode	Training / broadcast	Test accuracy
Direct inhibitory stream (input_mode=1)	LocalCA, per-soma rank-1	0.852 ± 0.007
Direct inhibitory stream (input_mode=1)	LocalCA, exact path transport	0.952 ± 0.000
Explicit I cells (input_mode=0)	LocalCA, path transport, I-cell updates on	0.942 ± 0.002
Explicit I cells (input_mode=0)	LocalCA, path transport, I-cell updates frozen	0.938 ± 0.002
Explicit I cells (input_mode=0)	standard backprop	0.951 ± 0.003

Table S4: **One-feedforward-layer input-mode path-gain probe on noise resilience (3 seeds).** All rows use one excitatory dendritic population with [3, 3] morphology and nonnegative inputs. The direct-I rows have no explicit inhibitory neurons; the input stream drives learned branch-level inhibitory conductance banks directly. The explicit-I rows use a matched inhibitory population with [3, 3] morphology. `local_ca` updates explicit inhibitory-cell dendrites with the same policy used for excitatory dendrites, while `freeze` holds those dendrites fixed but still lets excitatory I-to-E synapses learn. Exact path transport closes the direct-I rank-1 gap, and explicit inhibitory cells nearly match both direct-I path transport and the explicit-I backprop ceiling.

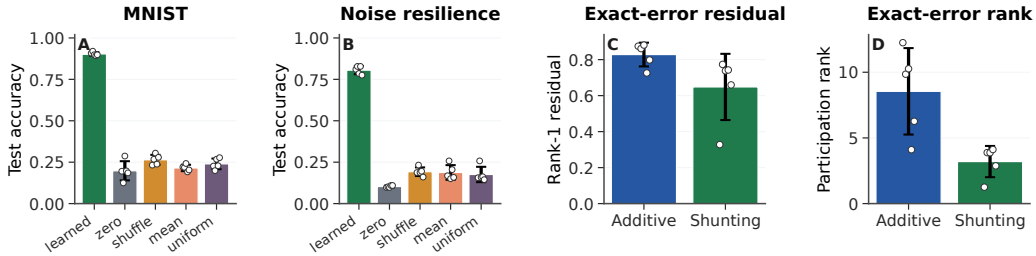


Figure S10: **Post-training inhibition intervention and exact-error compressibility.** (A,B) Post-training inhibitory-conductance interventions on trained shunting LocalCA checkpoints. Accuracy is measured on a 512-example held-out subset; dots show seeds. Removing, shuffling, or clamping learned inhibition collapses performance, showing that the learned inhibitory field carries sample-specific task structure. (C,D) SVD diagnostics of the exact compartment-error matrix on matched MNIST $N_I=5$ runs. Shunting has a smaller rank-1 residual and lower participation rank than additive, so the exact error field is more compressible. Error bars are s.d. across 5 seeds.

511 Cue-Routing Feedback-Rank Diagnostic

512 Cue routing tests the regime where one shared teaching signal is expected to fail: context determines
 513 which noisy cue is reliable, so different pathways should receive different credit. Figure S11 shows
 514 that the benchmark is learnable and that moving beyond rank-1 feedback helps, but the current
 515 path-structured variant does not yet beat the best unstructured low-rank setting. We therefore interpret
 516 this result as a failure-mode and rank-requirement diagnostic, not as a solved structured-feedback
 517 method.

518 Random Low-Rank Broadcast Channels

519 Frozen Activation-Choice Audit

520 B Implementation Details

521 Biological Plausibility Assumptions

522 The theorems and rules above rest on a small set of explicit assumptions. We state them as a checklist
 523 so that readers can see exactly what is assumed and where each assumption is used:

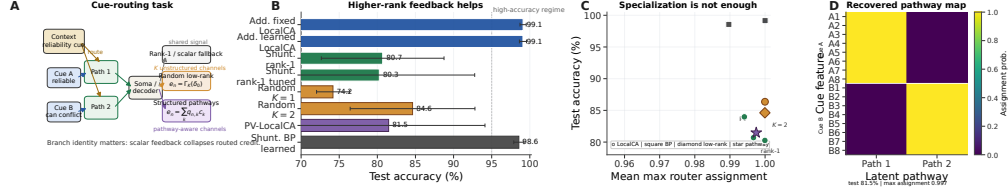


Figure S11: **Structured feedback when rank-1 broadcast fails.** (A) Cue-routing task schematic. Context determines which noisy cue is reliable on each trial, so different dendritic pathways should receive different teaching signals. (B) Five-seed comparison of rank-1, random low-rank, and pathway-structured feedback. Moving beyond rank-1 helps, while the best structured repair remains open. (C) Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success. (D) Recovered pathway assignments for a representative learned-router run. Cue-A and cue-B features segregate into separate latent pathways even when feedback quality, rather than router discreteness, is the limiting factor.

Setting	Broadcast	Test accuracy
<i>Noise resilience, shunting, rank bridge</i>		
per-soma	rank-1 per-soma	0.662 ± 0.016
path propagation	recursive attenuation proxy	0.762 ± 0.047
low-rank	$K=1$	0.453 ± 0.126
low-rank	$K=2$	0.630 ± 0.042
low-rank	$K=4$	0.764 ± 0.025
low-rank	$K=8$	0.795 ± 0.019
path transport	exact conductance-stage transport	0.847 ± 0.018
<i>Cue routing, learned shunting router, rank/structure sweep</i>		
per-soma	tuned rank-1	0.803 ± 0.125
low-rank	$K=1$	0.742 ± 0.022
low-rank	$K=2$	0.846 ± 0.082
low-rank	$K=4$	0.748 ± 0.085
pathway-vector	tuned structured rank-2	0.815 ± 0.126

Table S5: **Rank bridge and rank-structure comparison.** On shunting noise resilience, both the current path-propagation proxy and higher-rank random low-rank broadcast close a substantial fraction of the gap between rank-1 per-soma and exact path transport, with low-rank continuing to improve up to $K=8$. On cue routing, the learned-router sweep shows that moving beyond rank-1 helps, but the best current unstructured low-rank setting ($K=2$) is still slightly better than the tuned structured pathway-vector variant. We therefore interpret routed credit assignment as an open higher-rank problem rather than as a solved pathway-vector rescue.

- 524 **A1 (Steady-state voltage).** We use the steady-state solution of the passive cable equation (Eq. (1)).
525 Temporal dynamics are not modeled; all gradients, path gains, and fidelity measurements
526 refer to the conductance-stage voltage at steady state.
- 527 **A2 (Tree topology).** Each dendritic cell is a rooted tree with the soma as the root. There are no
528 recurrent connections within the dendrite, which makes the path from any compartment to
529 the soma unique and is what allows the closed-form path gain α_n in Theorem 1.
- 530 **A3 (Nonnegative conductances).** Synaptic and dendritic conductance parameters are pushed
531 through a positive transform (`softplus`), so $g_j^{\text{syn}}, g_j^{\text{den}} \geq 0$ at every optimisation step.
532 This is a biophysical constraint and is also what keeps the shunting denominator bounded
533 away from zero.
- 534 **A4 (Nonnegative presynaptic drive).** The first synaptic drive x is nonnegative, implemented either
535 by nonnegative inputs or an explicit ReLU transfer stage. Together with A3 this enforces
536 the positive-input condition used for all main sweeps.
- 537 **A5 (Pre-reactivation upward transfer).** Theorem 1 is stated for the pre-reactivation compartment
538 voltage. A post-voltage reactivation f (e.g. `param_tanh`) is handled by multiplying the

Dataset / strategy	Core	N_I	Identity (none)	Frozen param_tanh
MNIST / LocalCA	additive	0	0.780 ± 0.014	0.709 ± 0.037
MNIST / LocalCA	additive	10	0.907 ± 0.003	0.881 ± 0.005
MNIST / LocalCA	shunting	0	0.841 ± 0.007	0.778 ± 0.020
MNIST / LocalCA	shunting	10	0.905 ± 0.009	0.840 ± 0.016
Noise resilience / LocalCA	additive	0	0.590 ± 0.162	0.109 ± 0.006
Noise resilience / LocalCA	additive	10	0.859 ± 0.001	0.772 ± 0.009
Noise resilience / LocalCA	shunting	0	0.411 ± 0.028	0.331 ± 0.020
Noise resilience / LocalCA	shunting	10	0.769 ± 0.004	0.701 ± 0.006
MNIST / backprop	additive	0	0.925 ± 0.001	0.942 ± 0.001
MNIST / backprop	additive	10	0.921 ± 0.001	0.950 ± 0.001
MNIST / backprop	shunting	0	0.963 ± 0.0004	0.950 ± 0.001
MNIST / backprop	shunting	10	0.967 ± 0.001	0.946 ± 0.001
Noise resilience / backprop	additive	0	0.896 ± 0.003	0.106 ± 0.007
Noise resilience / backprop	additive	10	0.893 ± 0.002	0.927 ± 0.002
Noise resilience / backprop	shunting	0	0.898 ± 0.001	0.869 ± 0.002
Noise resilience / backprop	shunting	10	0.926 ± 0.001	0.912 ± 0.001

Table S6: **Frozen activation-shape audit on matched dendritic architectures (5 seeds per condition).** Identity (none) and a frozen bounded post-voltage reactivation param_tanh lead to markedly different operating regimes under nonnegative first-layer synaptic drive. In this audit, identity is the safer LocalCA choice on both MNIST and the noise-resilience task, while the backprop effect is strongly regime-dependent: frozen param_tanh helps additive most clearly in the inhibited noise-resilience setting ($N_I=10$) but collapses it at $N_I=0$. These runs match additive reactivation initialization to the shunting setting and hold the activation parameters fixed, isolating activation shape from both additive/shunting initialization differences and activation learning itself.

Dataset	Core	N_I	Identity (none)	Frozen param_tanh
MNIST	additive	0	CV 0.280, cosine 0.220	CV 1.122, cosine 0.074
MNIST	additive	10	CV 0.310, cosine 0.212	CV 1.079, cosine 0.094
MNIST	shunting	0	CV 0.393, cosine 0.170	CV 0.770, cosine 0.114
MNIST	shunting	10	CV 0.385, cosine 0.281	CV 1.177, cosine 0.106
Noise resilience	additive	0	CV 0.124, cosine 0.079	CV 0.114, cosine 0.052
Noise resilience	additive	10	CV 0.414, cosine 0.206	CV 0.840, cosine 0.094
Noise resilience	shunting	0	CV 0.255, cosine 0.324	CV 0.344, cosine 0.341
Noise resilience	shunting	10	CV 0.762, cosine 0.140	CV 2.497, cosine 0.079

Table S7: **Frozen activation-shape theory audit on matched LocalCA runs (5 seeds per condition).** Each entry reports conductance-stage path-gain dispersion (CV) and direct per-soma compartment-error fidelity (cosine between the per-soma broadcast and the exact pre-reactivation compartment error). The positive-input theory audit largely agrees with the empirical one: freezing param_tanh usually broadens path-gain dispersion and weakens per-soma fidelity, especially in the inhibited settings where the main mechanistic story lives. The one mild exception is low-inhibition shunting on noise resilience, where frozen param_tanh leaves fidelity roughly unchanged while still hurting final accuracy, suggesting that activation choice affects optimization and representation as well as direct credit fidelity.

539 path gain by $f'(V_n)$ along the recursion. The compartment error remains the sole non-local
540 quantity; only the effective path gain acquires local activation-derivative factors.

541 **A6 (Single perisomatic broadcast).** Every synapse receives one non-local scalar or low-rank vector
542 signal delivered from the soma; all other ingredients of the update rule (eligibility traces,
543 driving force, input resistance, covariance modulators) are computable from quantities
544 available at the synapse or compartment.

545 Each synapse therefore has access to: presynaptic activity x_j (local), compartment voltage V_n
546 (local membrane potential), reversal potential E_j (fixed by receptor type), and total input resistance

547 $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$ (computable from local conductances). The 4F modulator r_n^{4F} requires online
548 estimation of voltage covariance (biologically plausible via slow calcium signals); the 5F factor ϕ_n
549 requires a linear regression proxy (implementable via eligibility traces). The only non-local quantity
550 is the broadcast error e_n , which requires a top-down or neuromodulatory signal from the soma to
551 dendritic compartments. We assume per-neuron resolution (one error per output neuron) for the
552 per-soma broadcast mode.

553 Dendritic Structure in Code

554 We use the morphology notation $[b_1, b_2, \dots, b_D]$ for rooted dendritic trees, where each factor gives
555 the fan-in from one dendritic stage to the next on the path to the soma. For example, $[3, 3]$ means that
556 each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches,
557 for 9 distal leaves per soma. In the main feedforward models, synaptic inputs terminate on dendritic
558 branches rather than directly on the soma. The main setting uses direct input-driven inhibition: the
559 transfer layer creates nonnegative excitatory and inhibitory streams from the same external input, and
560 learned inhibitory conductance banks attach directly to excitatory dendritic branches. These models
561 therefore have branch-level inhibitory conductance but no separate inhibitory neuron population. The
562 explicit-I probe instead instantiates feedforward inhibitory cells, which transform excitatory activity
563 before projecting inhibitory conductance back onto excitatory dendrites. In that setting the inhibitory
564 cells are also dendritic modules and their synaptic conductances are trained by the same local rule
565 family. The additive and shunting cores are architecture-matched at the tree level; they differ in the
566 branch-level voltage integration rule rather than in whether a dendritic tree exists. In all reported
567 dendritic runs, synaptic and dendritic conductance parameters are pushed through positive transforms,
568 so conductances remain nonnegative; unconstrained decoder weights are separate readout parameters
569 and are not interpreted as conductances.

570 Post-Voltage Reactivation

571 The conductance stage produces a pre-reactivation compartment voltage V_n , after which each branch
572 layer may apply a monotone scalar nonlinearity. The main sweeps use a learnable bounded firing-rate
573 transform, $(\tanh(m(V - b)) + 1)/2$, together with a nonnegative transfer stage so that the first
574 synaptic drive remains nonnegative even after input corruption or structured task noise. Most main
575 feedforward sweeps use a linear decoder, while the CIFAR extension uses a nonlinear decoder and
576 therefore maps output error back to decoder-input/soma space through the decoder Jacobian before
577 applying LocalCA. Unless explicitly overridden, additive and shunting cores can start from different
578 reactivation initializations; activation-fairness ablations therefore match additive initialization to the
579 shunting setting when comparing alternative reactivation choices directly. The appendix activation
580 audit additionally fixes the bounded reactivation after initialization to isolate activation shape from
581 activation plasticity.

582 LocalCA uses occupancy-quantile initialization together with a calibration pass that floors tiny
583 quantile widths, clamps extreme fitted slopes, and reverts invalid per-layer fits rather than allowing
584 pathological intermediate values to propagate. In the non-somatic feedforward publication families,
585 this calibration leaves the results essentially unchanged relative to the same occupancy-quantile setup
586 without the additional validity checks: phase-1 capacity changes from 0.8786 to 0.8784 mean test
587 accuracy, shunting-regime sweeps from 0.5617 to 0.5641, morphology-scaling sweeps from 0.3600
588 to 0.3578, and error-shaping sweeps from 0.2860 to 0.2798. The calibration therefore improves
589 numerical safety without changing the qualitative LocalCA story.

590 Component Checks

591 A five-seed MNIST component check confirmed that the main LocalCA results are not driven
592 by a single calibration artifact. Removing the occupancy-quantile calibration changes shunting
593 accuracy negligibly (90.7% to 90.8%), removing learned reactivation parameters costs about 1 pp,
594 and removing the post-voltage reactivation is the largest single ablation (87.3%). Additive controls
595 show small gains from each calibration ingredient. We keep these checks as implementation support
596 rather than as a separate appendix figure.

Quantity	Symbol	Convention
Voltage	V	Conductance-stage shunting voltage in $[0, 1]$ under nonnegative drive; additive controls may be signed
Conductances	$g^{\text{syn}}, g^{\text{den}}$	Nonneg. via softplus
Leak conductance	g^{leak}	Set to 1
Input resistance	R^{tot}	≤ 1

Table S8: Units and normalization.

597 Units and Parameterization

598 Decoder Update Modes

599 W_{dec} maps $V_L \rightarrow \hat{y}$. Modes: **backprop** ($\nabla_W L$ via autograd), **local** ($\Delta W = \eta \langle \delta_0 V_L^\top \rangle_B$), **frozen**
600 ($\Delta W = 0$).

601 Representative Manuscript Architectures

602 Hyperparameters

603 Table S10 collects the core training hyperparameters for each main experiment. Learning rates are
604 applied through parameter groups (top- k , dendritic block-linear, reactivation, decoder) with the values
605 given in the representative config files; where a single LR is reported, every group uses that value.

606 Effective Broadcast Dimensionality in Main Architectures

607 Compute Resources

608 All experiments were run on an institutional GPU cluster. Most MNIST, Fashion-MNIST, and
609 synthetic-task runs used a single NVIDIA A100 or H100 GPU and finished in minutes to tens of
610 minutes per seed. The flattened CIFAR-10 runs required longer single-GPU jobs but still did not use
611 distributed or multi-GPU training. Reported main metrics are aggregated over 5 seeds unless noted
612 otherwise; selected morphology and input-mode probes use 3 seeds and are labeled as supportive
613 diagnostics.

614 Code, Data, and Asset Availability

615 MNIST, Fashion-MNIST, CIFAR-10, and MICrONS are public datasets or public benchmark assets
616 cited in the paper; we use them under their standard published access conditions and cite their original
617 sources in the bibliography. The present anonymous submission does not include a public code URL
618 in order to preserve double-blind review, but the paper specifies the model family, training modes, key
619 hyperparameters, and representative configurations, and we plan to release code and configuration
620 files after review. The paper does not release a new dataset or a stand-alone pretrained model asset.

621 Algorithm

Algorithm 1 Main 5F Rank-1 LocalCA Update

```

1: Input: Dendritic model, batch  $(x, y)$ , learning rate  $\eta$ 
2: Forward pass; loss  $L$ , output error  $\delta^y$ 
3: Somatic/core error  $\delta_0 = \text{decoder-input Jacobian product } J_{\text{dec}}(h_{\text{core}})^\top \delta^y$  (or  $W_{\text{dec}}^\top \delta^y$  for a linear decoder)
4: for each layer  $n$  (reverse) do
5:   Rank-1 broadcast  $e_n = \text{mean}(\delta_0) \mathbf{1}_n$ 
6:   Update EMA estimates for  $r_n^{4F}$  and  $\phi_n$  from batch voltages
7:    $\Delta g_j^{\text{syn}} \leftarrow \eta r_n^{4F} \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B$ 
8:    $\Delta g_j^{\text{den}} \leftarrow \eta r_n^{4F} \phi_n \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B$ 
9: end for
10: Optional controls replace line 4 with neuron-wise, rank- $K$ , path-structured, or transported-oracle broadcasts.
11: Clip gradients; optimizer step

```

622 C Theoretical Details

623 Random-Broadcast Alignment

624 **Proposition 3** (Supportive random-broadcast alignment). *Let the broadcast matrix B_n have i.i.d.*
625 *zero-mean entries with $\mathbb{E}[B_n^\top B_n] = \sigma_B^2 I$. Then the expected cosine between the resulting local*
626 *gradient and the exact gradient is positive, $\mathbb{E}[\cos \angle(g^{\text{local}}, g^{\text{exact}})] \geq c_n > 0$, by an argument*
627 *analogous to feedback alignment [9]. The constant c_n depends on how strongly local eligibility*
628 *factors correlate with the conductance-stage path gain in Eq. (4); shunting improves this correlation*
629 *by narrowing the spread of intermediate sensitivities.*

630 We treat this argument as supportive rather than central. The main theoretical object in the paper
631 is still the exact pre-activation compartment error $\partial L / \partial V_n$; under identity upward transfer this
632 reduces to $\alpha_n \delta_0$, while with post-voltage activation the corresponding effective path gain additionally
633 includes local $f'_n(V_n)$ factors. The main empirical question is how faithfully different broadcast fields
634 approximate that quantity.

635 Morphology-Aware Extensions

636 **Path-integrated propagation.** Modulate broadcast error by $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$, approxi-
637 mating depth attenuation from Eq. (4).

638 **Depth modulation.** Per-branch scaling $\kappa_j = \kappa_{\text{base}} / (d_j + d_0)$, mirroring cable attenuation.

639 **Dendritic normalization.** $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}} / (\sum_k g_k^{\text{den}} + \varepsilon)$, analogous to homeostatic scaling
640 [21].

641 **Pathway-vector feedback.** For tasks with latent pathway structure, the broadcast becomes a role
642 vector inferred from the router. Local pathway activity gates this vector. Upward block transport then
643 passes it to earlier branches, aligning their feedback with downstream descendants.

644 **Router-derived branch roles.** Each branch receives a role profile inferred from router assignments
645 and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then
646 modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-
647 consistent updates without imposing a heuristic branch taxonomy.

648 HSIC Auxiliary Objectives

649 Following [17], we use kernel-based HSIC objectives:

$$\mathcal{L}^{\text{self}} = B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Z \mathbf{H}), \quad \mathcal{L}^{\text{target}} = -B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Y \mathbf{H}).$$

Moderate weights (0.01–0.1) help on context gating; negligible on MNIST. Online statistics (r_n^{4F} , ϕ_n) use Welford’s algorithm [20].

Online Variant with Eligibility Traces

The continuous-time eligibility trace is

$$\tau_e \dot{e}_j^{\text{syn}} = -e_j^{\text{syn}} + x_j(E_j - V_n)R_n^{\text{tot}},$$

with update

$$\Delta g_j^{\text{syn}} \propto \int e_j^{\text{syn}}(t)m_n(t) dt$$

[18, 19].

D Secondary Weight Statistics

As a secondary biological check, we fit log-normal distributions to active excitatory and inhibitory synaptic weights after training on MNIST and Fashion-MNIST. This analysis is descriptive rather than central to the credit-assignment claim. The main qualitative pattern is that shunting produces narrower and more stable weight distributions than additive integration, with the shunting conditions falling closer to descriptive biological reference widths from Song et al. [38] and MICrONS [40].

The same self-normalizing denominator that shapes credit also provides a plausible explanation for this secondary trend: larger total conductance lowers R_n^{tot} , reducing future update magnitudes on already high-conductance branches. This resembles multiplicative synaptic dynamics [41], but we do not use weight-distribution fits as evidence for the main credit-assignment claims.

E Task and Robustness Details

Each synthetic task is generated procedurally from a fixed random seed and uses fixed train/val/test splits for reproducibility. The MNIST-derived synthetic tasks use flattened 28×28 inputs; cue integration instead uses a small vector input with two output classes. Output dimensionality therefore varies by task and is stated in the task description when it differs from 10-way digit classification.

Context gating. Inputs are contextual figure-ground MNIST examples. The right half of the image carries the original MNIST digit, while the left half is replaced by low-amplitude uniform noise; all pixels therefore remain nonnegative. This task tests whether credit assignment can exploit a simple context-dependent spatial structure without leaving the nonnegative-drive regime of the conductance model. The HSIC auxiliary objective (weight 0.01) is used to decorrelate layer representations, adding roughly 3 pp under local learning.

Noise resilience. Inputs are flattened MNIST images corrupted by structured channel noise: a fixed 50-dimensional latent Gaussian perturbation is projected into input space and added to each example, with $\sigma_{\text{task}} = 1.5$, then clipped to $[0, 1]$. The projection is fixed across training, so the network must filter correlated interference to recover the underlying class signal. Credit assignment must remain coherent under the corrupted signal, making this task particularly sensitive to path-gain dispersion.

Info shunting. Inputs contain two additive streams: a low-frequency signal and a high-frequency distractor drawn with equal energy, together with a separate context channel indicating which stream is relevant on each trial. This benchmark uses signed synthetic inputs, so it violates the nonnegative presynaptic-drive assumption enforced in the main experiments. We therefore retain it only as an exploratory stress test and exclude it from the main-text claims.

Cue integration. Two noisy cue streams (A and B) are presented simultaneously, each carrying partial information about one of 2 classes. A binary one-hot context signal indicates which cue is reliable on each trial (cue A or cue B), and each cue vector is clipped to $[0, 1]$ after noise is added. The fixed-pathway variant duplicates the context into both cue branches; the learned-routing variant requires the router to discover the cue separation from data. This task is used in Sec. A to test structured pathway-vector broadcast while staying inside the nonnegative-input regime.

Depth Scaling and Noise Robustness

Depth scaling. Varying dendritic depth from 1–4 layers (branch factors [9] to [3, 3, 3, 3]): shunting local degrades from 63.5% to 57.4%; additive local degrades more steeply from 54.9% to 29.7%. The shunting advantage grows with depth (+8.5 → +27.7 pp). Backprop ceilings remain stable at about 90–92%. See Fig. S5A.

Noise robustness. Gaussian noise $\mathcal{N}(0, \sigma^2)$ on broadcast error: shunting is robust to $\sigma \leq 0.1$ (about 62%) while additive drops from 46.5% to chance at $\sigma=1.0$, confirming that shunting credit signals carry useful learning information beyond the broadcast magnitude. See Fig. S5B.

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743 and how they scale with dataset size.
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753 Question: For each theoretical result, does the paper provide the full set of assumptions and
754 a complete (and correct) proof?

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758 cross-referenced in the main text.

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773 of the paper (regardless of whether the code and data are provided or not)?

774 Answer: [Yes]

775 Justification: The main paper and appendices specify the datasets, architectures, local-rule
776 family, broadcast modes, representative hyperparameters, and compute regime needed to
777 reproduce the headline results; see Sec. 4 and Appendix B.

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786 For example, if the contribution is a novel architecture, describing the architecture fully
787 might suffice, or if the contribution is a specific model and empirical evaluation, it may
788 be necessary to either make it possible for others to replicate the model with the same
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790 one good way to accomplish this, but reproducibility can also be provided via detailed
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Setting	Encoder	E layers	Tree	Synapses / branch	Seeds	Notes
Gradient fidelity / path-gain / noise-resilience mechanism	id.+ReLU	[128, 128]	[3, 3]	$N_E=40, N_I \in \{0, 5, 10, 20, 40\}$	5 fidelity, 5 oracle	Main mechanism sweeps in Figs. 2 and 3; reactivation <code>param_tanh</code> .
MNIST / context-gating local competence	id.+ReLU	[128]	[3, 3]	$N_E=40, N_I=20$	5	Best local 5F per-soma runs in Table S2; reactivation <code>param_tanh</code> .
Fashion-MNIST competence	id.+ReLU	[128]	[3, 3]	$N_E=40, N_I=20$	5	Dedicated five-seed standard-vs-local competence sweep; reactivation <code>param_tanh</code> .
Cue-routing PV-LocalCA	router	[64]	[2]	$N_E=10, N_I=4$	5	Learned router with two pathway groups and rank-2 structured feedback; reactivation <code>param_tanh</code> .
CIFAR-10 compact direct-I-stream control ladder	id.+ReLU	[20] E, no I cells	[3, 3, 3, 3]	$N_E=25, N_I=25$ direct I-to-E, with matched no-I controls	5 per condition	Harder-data extension with input-driven inhibitory conductance, decoder [32, 16], decoder-aware soma mapping, and additive fairness controls.
Morphology $\times$ inhibition appendix map	id.+ReLU	[128, 128]	[2, 2], [3, 3], [4, 4], [2, 2, 2], [3, 3, 3]	$N_E=40, N_I \in \{0, 5, 10, 20, 40\}$	3	Supportive 3-seed sweep; reactivation <code>param_tanh</code> .

Table S9: Representative architectures used in the manuscript. All dendritic rows use explicit branch-level excitatory and inhibitory synapse banks with `somatic_synapses=false`, nonnegative conductances, and a nonnegative first-layer drive enforced by an explicit ReLU transfer stage in the main runs. The tree column gives dendritic morphology per excitatory neuron, and the synapse counts report excitatory and inhibitory synapses per branch.

Setting	Opt.	LR	Batch	Epochs	W-dec.	Notes
MNIST / Fashion-MNIST / context gating (5F per-soma LocalCA)	Adam	$10^{-3} / 5 \cdot 10^{-4}$ (block / react.)	256	100	0	fixed-epoch, no early stopping; <code>param_tanh</code> reactivation
MNIST / Fashion-MNIST / context gating (BP ceiling)	Adam	$10^{-3} / 5 \cdot 10^{-4}$	256	100	0	matched to LocalCA setup
Gradient fidelity / path-gain / noise resilience	Adam	10^{-3}	256	50	0	5 seeds, hooks enabled for diagnostic capture
Morphology $\times$ inhibition regime map (exploratory)	Adam	10^{-3}	256	50	0	3 seeds
Cue routing	Adam	10^{-3}	256	90	0	early stopping with patience 30
CIFAR-10 compact direct-I-stream depth-4 LocalCA	Adam	$10^{-3} / 10^{-4}$ (block / react.)	256	400	0 / 0.01	grad-clip 5.0, early stop patience 50, nonlinear decoder
CIFAR-10 compact direct-I-stream depth-4 BP	Adam	10^{-3}	256	200	0.01	early stop patience 40, matched ceiling
Weight-statistics summary	Adam	10^{-3}	256	100	0	<code>weight_analysis</code> hook captures final-epoch statistics

Table S10: Training hyperparameters by experiment. All runs use Adam with default $\beta_1=0.9$, $\beta_2=0.999$, $\varepsilon=10^{-8}$. Reactivation and block-linear parameter groups use a slower LR to keep conductance-level dynamics stable. Full configs are provided under `drafts/dendritic-local-learning/configs/`.

Setting	Hidden width(s)	Output dim	Effective default per-soma field
Gradient fidelity / path-gain / noise resilience	128, 128	10	Scalar / rank-1 at both dendritic layers
MNIST / Fashion-MNIST competence	128	10	Scalar / rank-1 at the dendritic hidden layer
Context gating competence	128	2	Scalar / rank-1 at the dendritic hidden layer
Cue routing	64	2	Scalar / rank-1 by default; higher-rank tests use explicit low-rank or pathway-vector feedback
Compact CIFAR-10 harder-data extension	20	decoder-input dim 20	Vector-valued per-soma after decoder-aware mapping; higher-rank tests use explicit low-rank or path transport

Table S11: Effective dimensionality of the default neuron-wise broadcast in the main architectures. The implementation uses the output error vector only when layer width matches decoder output dimension; otherwise it falls back to a shared scalar. The main feedforward settings in this paper therefore test a genuinely rank-1 broadcast regime by default, which is why the higher-rank low-rank, path-transport, and pathway-vector controls are important.

Asset	Use in this paper	Access / license note
MNIST	Digit classification and derived nonnegative synthetic tasks	Public academic benchmark credited to the original dataset authors; used through standard public access, with no raw-data redistribution.
Fashion-MNIST	Apparel classification stress test	Public benchmark released with the Fashion-MNIST paper and repository; used through standard public access, with no raw-data redistribution.
CIFAR-10	Flattened harder-data diagnostic	Public research benchmark from the CIFAR dataset collection; used through standard public access, with no raw-data redistribution.
MICrONS	Biological reference statistics in the weight-distribution appendix	Public connectomics resource cited to the MICrONS Consortium; only aggregate comparison statistics are reported here.
Synthetic tasks	Context gating, noise resilience, cue integration	Generated procedurally from public benchmarks or random seeds described in Appendix E; no new third-party asset is introduced.

Table S12: Existing assets used in the manuscript. We credit the original sources in the bibliography, use public benchmark assets under their published access conditions, and do not redistribute third-party raw data in the anonymous submission.

Condition	MNIST		Fashion-MNIST	
	Exc σ	Inh σ	Exc σ	Inh σ
BP + Shunting	0.94 $\pm$ 0.01	0.87 $\pm$ 0.01	0.92 $\pm$ 0.01	0.84 $\pm$ 0.01
BP + Additive	1.30 $\pm$ 0.01	1.22 $\pm$ 0.01	1.16 $\pm$ 0.01	1.05 $\pm$ 0.01
Local + Shunting	1.22 $\pm$ 0.01	1.71 $\pm$ 0.02	1.13 $\pm$ 0.06	1.36 $\pm$ 0.06
Local + Additive	1.69 $\pm$ 0.16	1.57 $\pm$ 0.07	1.43 $\pm$ 0.16	1.11 $\pm$ 0.12
Song et al. 2005	$\sigma = 0.94$ (rat V1 L5, EPSP amplitude)			
MICrONS E $\rightarrow$ E	$\sigma = 1.14$ (mouse V1, synapse size, $n=3.9M$)			
MICrONS I $\rightarrow$ E	$\sigma = 0.92$ (mouse V1, synapse size, $n=3.5M$)			

Table S13: **Secondary weight-distribution summary.** Log-normal σ of excitatory and inhibitory synaptic weights (mean $\pm$ std across 5 seeds). Biological references are descriptive anchors only. With biologically constrained E/I ratios (4:1), shunting self-normalization produces weight heterogeneity in the range spanned by those references, whereas additive integration is broader and less stable.